

Math 527, S10
Exam 1

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	25	
2	25	
3	25	
4	25	
Total	100	

Good Luck !

Problem 1. (25 points)

(a) A real number x is represented approximately by 1.5072, and we are told that the relative error is at most 0.02%. What is x ?

$$|error| = |\tilde{x}| \cdot |relative\ error| \quad (\text{with } \tilde{x} = 1.5072) \\ = 1.5072 \cdot 0.0002$$

$$\Rightarrow \tilde{x} - |error| \leq x \leq \tilde{x} + |error|$$

$$1.5072 \cdot (1 - 0.0002) \leq x \leq 1.5072 \cdot (1 + 0.0002)$$

$$1.50689856 \leq x \leq 1.50750144$$

(b) The Taylor series for $\ln(1+x)$ state that

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{x^k}{k}.$$

Use this form to compute $\ln(1.0001)$ correct to 14 decimal places (rounded).

$$1+x = 1.0001$$

$$\Rightarrow x = 0.0001 = 10^{-4}$$

$$\frac{x^4}{4} = \frac{10^{-16}}{4} < 10^{-14}$$

Thus, we could get

$$\ln(1.0001) \approx (0.0001) - \frac{(0.0001)^2}{2} + \frac{(0.0001)^3}{3}$$

$$\approx 0.00009999500033$$

Problem 2. (25 points) Two of the four zeros of $f(x) = x^4 + 2x^3 - 7x^2 + 3$ are positive.
 (a) Write down Newton's method in simplified form for determining the roots of the above $f(x)$?

$$X_{k+1} = X_k - \frac{f(X_k)}{f'(X_k)}$$

$$\Rightarrow X_{k+1} = X_k - \frac{X_k^4 + 2X_k^3 - 7X_k^2 + 3}{4X_k^3 + 6X_k^2 - 14X_k}$$

$$k = 0, 1, 2, \dots$$

(b) Find these two positive roots by Newton's method with $x_0 = 1.0$ and $x_0 = 2.0$, respectively.
 (Run two steps for each of them)

When $x_0 = 1.0$, we have

$$x_1 = 0.75$$

$$x_2 \approx 0.79094828$$

When $x_0 = 2.0$, we have

$$x_1 = 1.75$$

$$x_2 \approx 1.64158163$$

Problem 3. (25 points)

(a) Write down the iterative formula when we use the secant method on $f(x) = 2 - e^x$. Compute x_2 with $x_0 = 0$ and $x_1 = 1$.

$$\begin{aligned} x_{k+1} &= x_k - \left(\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})} \right) f(x_k) \\ &= x_k - \left(\frac{x_k - x_{k-1}}{(2 - e^{x_k}) - (2 - e^{x_{k-1}})} \right) (2 - e^{x_k}) \\ &= x_k + \frac{(x_k - x_{k-1})(2 - e^{x_k})}{e^{x_k} - e^{x_{k-1}}} \end{aligned}$$

With $x_0 = 0$, $x_1 = 1$, we have

$$x_2 \approx 0.58197671$$

(b) Use the Lagrange interpolation process to obtain a polynomial of least degree that assumes these values:

x	1	2	4
y	6	7	15

$$l_0(x) = \left(\frac{x - x_1}{x_0 - x_1} \right) \left(\frac{x - x_2}{x_0 - x_2} \right) = \left(\frac{x - 2}{1 - 2} \right) \left(\frac{x - 4}{1 - 4} \right) = \frac{1}{3} (x - 2)(x - 4)$$

$$l_1(x) = \left(\frac{x - x_0}{x_1 - x_0} \right) \left(\frac{x - x_2}{x_1 - x_2} \right) = \left(\frac{x - 1}{2 - 1} \right) \left(\frac{x - 4}{2 - 4} \right) = -\frac{1}{2} (x - 1)(x - 4)$$

$$l_2(x) = \left(\frac{x - x_0}{x_2 - x_0} \right) \left(\frac{x - x_1}{x_2 - x_1} \right) = \left(\frac{x - 1}{4 - 1} \right) \left(\frac{x - 2}{4 - 2} \right) = \frac{1}{6} (x - 1)(x - 2)$$

$$\begin{aligned} \Rightarrow P(x) &= y_0 l_0(x) + y_1 l_1(x) + y_2 l_2(x) \\ &= 2(x - 2)(x - 4) - \frac{7}{2} (x - 1)(x - 4) + \frac{5}{2} (x - 1)(x - 2) \\ &= x^2 - 2x + 7 \end{aligned}$$

Problem 4. (25 points)

(a) Complete the following divided-difference table:

x	$f[\]$	$f[\ ,\]$	$f[\ ,\ ,\]$	$f[\ ,\ ,\ ,\]$	$f[\ ,\ ,\ ,\ ,\]$
0	1				
1	9	8			
2	23	14	3		
4	93	35	7	1	
6	259	83	12	1	0

(b) Using the above results to construct a Newton's interpolation polynomial that assumes these values.

$$\begin{aligned}
 P(x) &= a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2) \\
 &\quad + a_4(x-x_0)(x-x_1)(x-x_2)(x-x_3) \\
 &= 1 + 8x + 3x(x-1) + x(x-1)(x-2)
 \end{aligned}$$

(c) Write the polynomial obtained above in nested form for easy evaluation.

$$P(x) = 1 + x(8 + (x-1)(3 + (x-2)))$$

Extra Credit. (5 points)

Determine the error term for the approximation formula:

$$f'(x) = \frac{1}{4h} [f(x+3h) - f(x-h)].$$

$$f(x+3h) = f(x) + 3hf'(x) + \frac{(3h)^2}{2} f''(\xi_1) \quad \xi_1 \text{ between } x \text{ and } x+3h$$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2} f''(\xi_2) \quad \xi_2 \text{ between } x \text{ and } x-h$$

$$\Rightarrow \frac{1}{4h} [f(x+3h) - f(x-h)]$$

$$= \frac{1}{4h} \left[4hf'(x) + h^2 \left(\frac{9}{2} f''(\xi_1) - \frac{1}{2} f''(\xi_2) \right) \right]$$

$$= f'(x) + h \left(\frac{9}{8} f''(\xi_1) - \frac{1}{8} f''(\xi_2) \right)$$

$$= f'(x) + hf''(\xi) \quad \text{for some } \xi \text{ between } x-h \text{ and } x+3h$$

$$\Rightarrow \text{error term } hf''(\xi) = O(h)$$